

Leo Ng

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Profile

- Bachelor of Engineering (Electrical) and Computer Science student at UNSW with a Distinction average
- Hands-on experience across electronics, embedded systems, PCB design, programming, automation and hardware/software integration

Education

Bachelor of Engineering (Electrical) and Computer Science
UNSW Sydney

2023 – Expected Aug 2027

- Achieved a Distinction average

Engineering Projects

Embedded Sensing System / Piano Teaching Aid
(UNSW Course Engineering Project)

May 2026 – Aug 2026

- Designed and built an RP2040-based piano teaching aid using 13 VL53L1X ToF sensors for real-time key and hand-position detection
- Integrated multiple I²C sensors using XSHUT address control and debugged startup conflicts, timeouts and inconsistent readings
- Designed custom PCB schematics and layouts in KiCad, including sensor interfaces, power regulation, connectors and LED circuitry

Technologies: RP2040, C/MicroPython, VL53L1X, I²C, KiCad, PCB Design

Custom Flip Phone – Hardware & Software Integration
(Personal Engineering Project)

Aug 2026 – Present

- Developing a compact flip-phone integrating an Orange Pi 3B, MIPI DSI display, keypad controls and Android software
- Investigated display pinouts, MIPI DSI configuration, voltage requirements and Linux/Android panel-driver integration for hardware bring-up

Technologies: Orange Pi, MIPI DSI, Android, Kotlin, Linux

AI Solutions / Workforce Scheduling Automation
(UNSW Industry Project)

Aug 2024 – Sep 2024

- Analysed Excel-based rostering inefficiencies for Gate Gourmet Japan and evaluated automation opportunities
- Developed a Python prototype to automate roster-processing workflows and presented implementation recommendations

Technologies: Python, Excel, Automation, Technical Analysis

Employment

Freelance Website Developer

Real Estate Company

Apr 2024 – Nov 2024

- Developed and customised a client-facing website for a real estate company using Wix and Wix Velo
- Implemented responsive pages, custom JavaScript functionality and backend integrations to support business and client workflows

Technologies: Wix, Wix Velo, JavaScript, CSS, Web Development

Restaurant Team Member & Operations Support

Pho Sam Lane Cove

Dec 2022 – Current

- Support stocktake, inventory tracking, supplier invoicing and day-to-day workflow coordination
- Troubleshoot POS systems, devices and other technology issues to support reliable restaurant operations

Technical Skills

- **Programming:** C, Python, JavaScript, Shell, Kotlin
- **Electrical & Embedded:** RP2040, I²C, sensor integration, PCB design, power regulation, circuit testing
- **Engineering Tools:** KiCad, MATLAB, Simulink, LTspice, Git, Linux
- **Engineering Practice:** debugging, prototyping, component selection, validation, hardware/software integration

Languages

- English – Fluent
- Cantonese – Native
- Mandarin – Conversational
- Japanese – Basic
- German – Basic (A2 Level)